MindPal - AI Therapeutic Support System

Measurement theory application Summary

1. METROLOGY

MindPal applies metrology through standardized measurement instruments built into the system. A 1-10 mood scale captures subjective emotional state at the start of each session, while a session timer records engagement duration in seconds.

These tools produce consistent, reproducible measurements across all users and devices, aligned to clinical standards used in therapeutic practice.

2. PROPERTY-ORIENTED MEASUREMENT

Each of MindPal's nine therapy modes targets a distinct, well-defined therapeutic property. CBT mode measures cognitive distortion frequency,DBT mode measures distress tolerance levels, Anxiety mode measures anxiety severity using GAD-7 aligned questioning, and Depression mode measures low mood severity using PHQ-9 aligned questioning. This ensures each measurement is focused on one specific attribute rather than a vague general concept of wellness.

3. MEANINGFULNESS IN MEASUREMENT

All measurements in MindPal carry real clinical significance. The 1-10 mood scale maps directly to recognized levels of psychological distress used by practicing therapists. When a user scores very low, the system triggers crisis-aware responses and provides validated crisis resources (988 Suicide and Crisis Lifeline, Crisis Text Line 741741). Every measurement threshold produces a meaningful, proportionate clinical response rather than a generic reply.

4. SCALE

MindPal uses four scale types across its measurement instruments. The therapy mode selector uses a nominal scale of categories with no numerical order (CBT, DBT, Mindfulness, etc.). The mood check-in uses an ordinal scale of ordered values from 1 to 10. The session timer uses an interval scale of equal time units with meaningful differences.

Message count per session uses a ratio scale of equal intervals with true zero, meaning no messages equals no session.

5. MEASUREMENT VALIDATION

MindPal validates its measurements at three levels. Content validity is achieved by grounding every therapy mode prompt in peer-reviewed clinical literature of CBT prompts reference Beck's cognitive model, DBT prompts reference Lineman's four-module framework. Construct validity is maintained by aligning the mood scale to the well-validated construct of subjective emotional wellbeing. Reliability is supported by consistent use of the same AI model (Llama 3.3 70B via Groq), the same temperature setting (0.75), and server-side input validation on every request.

6. OBJECT-ORIENTED MEASUREMENT

MindPal measures the system as a collection of interacting objects. The Session object tracks duration, message count, and mode selected. The Message object tracks role (user or assistant) and content length. The Mode objects tracks which therapy specialization is active. The response object tracks reply length, latency, and error state. The conversation history array in the frontend represents the full measurable state of each therapeutic session at any moment in time.

7. SUBJECT-DOMAIN-ORIENTED MEASUREMENT

Every measurement instrument in MindPal is designed specifically for the mental health domain. The nine therapy modes correspond to the nine most widely practiced therapeutic modalities in clinical settings. Cognitive distortions are classified using Beck's established taxonomy.DBT skills are structured using Linehan's own four modules. Grief responses follow Worden's Tasks model. Trauma measurement follows the five trauma-informed care standards of safety, trustworthiness, choice, collaboration, and empowerment. The system is built on domain expertise, not general-purpose design.